MGU-UGP (HONOURS)

FOURTH SEMESTER MODEL EXAMINATION

(2024 ADMISSION ONWARDS)

MG4SECCND200– FOOD ADULTERATION SCHEME

Duration: 2hrs Maximum Marks: 50

Part A

Multiple Choice Questions

Answer All Questions

Each question carries 1 mark

1. Pesticide residues present in fruits
2. Metanil Yellow
3. Artificial colouring agent
4. Iodine test
5. Monosodium glutamate
6. Baking powder

[1 × 6 = 6 Marks]

Part B

Short Answer Type Questions

Answer 2 Questions

Each question carries 2 marks

7. Define food adulteration. (2 marks)

- Food adulteration is the intentional or unintentional addition of inferior or harmful substances to food or removal of valuable components. *(1 mark)*
- It reduces food quality, safety, and nutritional value. *(1 mark)*

8. List the three types of food hazards. (2 marks)

- Biological hazards – microorganisms such as bacteria, viruses, fungi. *(0.5)*
- Chemical hazards – pesticides, toxins, heavy metals. *(0.5)*
- Physical hazards – foreign objects like glass, metal, stones. *(0.5)*

- Correct listing of hazards. *(0.5)*

9. Difference between additive and adulterant. (2 marks)

Additive	Adulterant
Substance intentionally added to improve quality, shelf life or appearance	Substance added illegally to reduce cost or increase quantity

Marking:

- Correct definition of additive – 1 mark
- Correct definition of adulterant – 1 mark

10. Define food colourants (2 marks)

- Food colourants are substances added to food to improve or restore colour. *(1 mark)*
- They enhance appearance and consumer acceptance. *(1 mark)*

[2 × 2 = 4 Marks]

Part C

Short Essay Type Questions

Answer **4** Questions

Each question carries **5** marks

11. Health hazards caused by adulterants (5 marks)

- Toxic effects *(1)*
- Digestive disorders *(1)*
- Allergic reactions *(1)*
- Organ damage (liver/kidney) *(1)*
- Carcinogenic effects or chronic diseases *(1)*

12. Steps in HACCP (5 marks)

1. Conduct hazard analysis *(1)*
2. Identify critical control points (CCPs) *(1)*

3. Establish critical limits *(1)*
4. Monitoring procedures *(1)*
5. Corrective actions *(0.5)*
6. Verification procedures *(0.5)*

13. Role of emulsifiers in ice cream (5 marks)

- Stabilize fat and water mixture *(1)*
- Improve texture *(1)*
- Increase whipping ability *(1)*
- Prevent ice crystal formation *(1)*
- Provide smooth and creamy structure *(1)*

14. Types of preservatives (5 marks)
Natural preservatives

- Salt *(1)*
- Sugar *(1)*
- Vinegar *(1)*

Chemical preservatives

- Sodium benzoate *(1)*
- Potassium sorbate / sulphites *(1)*

15. Natural vs Artificial sweeteners (5 marks)
Natural sweeteners

- Honey *(1)*
- Stevia *(1)*

Artificial sweeteners

- Aspartame *(1)*
- Sucralose *(1)*

Comparison

- Artificial sweeteners have high sweetness with low calories *(1)*

16. Permitted and Non-permitted colours (5 marks)
Permitted colours

- Tartrazine *(1)*
- Sunset Yellow *(1)*
- Carmoisine *(1)*

Non-permitted colours

- Metanil yellow *(1)*
- Rhodamine B *(1)*

[5 × 4 = 20 Marks]

Part D

Essay Type Questions

Answer **2** Questions

Each question carries **10** marks

17. Biological, Physical and Chemical hazards (10 marks)
Biological hazards (3 marks)

- Bacteria (Salmonella, E.coli) *(1)*
- Viruses *(1)*
- Fungi / parasites *(1)*

Physical hazards (3 marks)

- Glass pieces *(1)*
- Metal fragments *(1)*
- Stones, hair, plastic *(1)*

Chemical hazards (4 marks)

- Pesticide residues *(1)*
- Heavy metals *(1)*
- Food additives misuse *(1)*
- Cleaning chemicals / toxins *(1)*

18. Functions of FSSAI (10 marks)
    - Formulate food safety standards *(2)*
    - Regulate manufacture, storage and sale of food *(2)*
    - Issue licenses to food businesses *(2)*
    - Ensure food quality and safety *(2)*
    - Promote food safety awareness and research *(2)*
19. Role of hydrocolloids in texture (10 marks)
    - Increase viscosity *(2)*
    - Stabilize emulsions *(2)*
    - Control moisture *(2)*
    - Improve mouthfeel *(2)*
    - Examples: pectin, guar gum, xanthan gum *(2)*

20. Relative sweetness of sugars (10 marks)
Definition

- Relative sweetness is the sweetness level of a substance compared with sucrose (standard value = 100). *(3 marks)*

Comparison

- Fructose – sweetest *(2)*
- Sucrose – reference sugar *(2)*
- Glucose – moderately sweet *(1.5)*
- Maltose – least sweet *(1.5)*

[10 × 2 = 20 Marks]